Enhancing nutrient recovery and compost maturity of coconut husk by vermicomposting technology

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Highlights

- Earthworm could convert coconut husk mixed with animal manure into a valuable compost.
- Mixing of animal manure with coconut husk increased the relative nutrient recovery.
- Index value for maturity parameters were derived for composting of coconut husk.
- Manurial value of coconut husk compost improved by pig slurry substitution.

Abstract

Vermicompost was prepared by five different treatments from relatively resistant coconut husk mixed with either pig slurry or poultry manure. The recovery of vermicompost varied from 35% to 43% and it resulted in significant increase in pH, microbial biomass carbon, macro and micro nutrients concentration. Among the treatments highest relative N (1.6) and K (1.3) recovery were observed for 20% feedstock substitution by pig slurry while poultry manure substitution recorded highest P recovery (2.4). Compost maturity parameters significantly differed and well correlated. The characteristics of different treatments established the maturity indices as C/N 15–20; Cw < 1.8; Cw/Norg < 0.55; Lignin < 10–12; CHA/CFA > 1.5 and HI > 15.0. The manurial value of the coconut husk compost was improved by feedstock substitution with pig slurry (80:20). The results

revealed the technical feasibility of converting coconut husk into valuable compost by feedstock substitution with pig slurry.

Introduction

Coconut is produced in 92 countries on about 11.8 million hectares of land with estimated production of 61.7 million tons (FAO, 2009). In India it is produced in all the coastal, north-eastern and island region in an area of 2.14 million hectares with the total production of 21.67 billion nuts (CDB, 2014). In Andaman Nicobar Islands, coconut is the major crop occupying 21,900 ha with a production of 130 million nuts. Coconut husk is the major waste generated from coconut industry with global estimated production of 23 million tons (FAO, 2009). Though the coconut husk is used in many ways like fuel, mulch, extraction of coir fiber, coco peat etc, in many countries it is dumped as a waste causing major environmental concern. In addition, this amounts to wastage of several thousand tons of plant nutrients which are locked in this organic waste. These concerns can be addressed by recycling the wastes through proper composting.

Vermicomposting is considered as one of the alternative options for transforming coconut wastes into nutrient rich compost useful for plants and soil while diminishing their negative environmental impact. Although microbes are responsible for the biochemical degradation of organic matter, earthworms are the important drivers of the process by conditioning the substrate and altering the biological activity. Many studies have been reported on the vermicomposting of agro-wastes (Bansal and Kapoor, 2000, Hanc and Chadimova, 2014), coconut leaves, coir pith and cow dung (Gopal et al., 2009); coconut flakes and coconut husk with goat manure (Tahir and Hamid, 2012) into a nutrient rich manure. But it was widely accepted that biodegradation of coconut husk is difficult and the recovery of nutrients are low as it has higher lignocellulosic material compared to other agro-wastes. This necessitates the addition of other nitrogen rich animal wastes such as cow dung, poultry manure, goat manure, pig slurry and standardization of feedstock composition and determination of maturity of the compost.

This is very essential because the principal requirement of compost for its safe use in soil is the degree of stability and maturity since phytotoxic compounds are produced by the microorganisms in unstable composts (Zucconi et al., 1985). In general, organic carbon content, degree of organic matter humification, water soluble carbon, cation exchange capacity, the ratios of C/N, water soluble C/organic N, humic acid to fulvic acid and amount of CO_2 evolved were used to describe the compost maturity (Bernal et al., 1998). In addition the recovery of nutrients from feedstock mixtures used in a particular composting method in comparison to initial as well as control reveals the efficiency and acceptability of the method. This is more pertinent to lignified and slowly degradable organic wastes like coconut husk.

Nicobar group of islands located in the Indian Ocean region has been declared as area for organic farming which generated huge demand for organic source of nutrients prepared using locally available materials. Piggery and backyard poultry other than coconut plantations are the major farm activities found in many of the tropical islands as in the case of Nicobar Islands (Swarnam et al., 2015). There is ample scope to utilize these animal wastes for recycling the nutrients locked in the coconut husk within the farms. In view of these problems and opportunities this study was carriedout to investigate the feasibility of vermicomposting the coconut husk with either pig slurry or poultry manure in suitable proportion for agricultural use. Further due to the resistant nature of coconut husk to decomposition and long composting period attempts were also made to derive index values for some of the compost maturity parameters to facilitate evaluation of the efficiency of different composting methods and compost quality.

Feedstock and treatment details

Coconut husk, poultry manure and pig slurry obtained from Farming System Research Unit of the institute were used as feedstocks. The coconut husk was shredded into small pieces of 1–2 cm to increase the surface area for liquid adhesion and microbial contact. The shredded material was soaked in water for a week to improve the absorption capacity for facilitating faster decomposition. The water used for retting the coconut husk is used for sprinkling the compost pile to avoid any nutrient loss.

Compost recovery

The final compost produced from the waste material is essential for planning the amount of feedstock required in a given area and space. In the present study weight and volume reduction by and large were uniform with no significant difference among the treatments (T1–T4) but varied significantly when compared to coconut husk alone (T5) which was used as control. The final recovery of the compost at the end of 120 days composting period varied from 35.0% to 42.5% of the initial feed stock which

Conclusions

Vermicomposting of coconut husk either with pig slurry or poultry manure recycles coconut waste. The nutrient content of the final product significantly increased in all the treatments. The addition of pig slurry at 80:20 ratio recorded highest nutrient recovery of both total and available forms with favorable compost maturity and quality parameters. The fertilizing index of the final compost for different treatments varied from 4.0 to 4.7 indicating high manurial value and its potential use

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